PG No. 13 C_{4v} $4mm$ [tetragonal]

Table 1: Representation matrices

Irrep.						
A_1	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$				
A_2	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$				
B_1	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$				
B_2	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$				
E	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}0&-1\\-1&0\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}0&1\\1&0\end{bmatrix}$				